

Human–AI Interaction and the Internalization of Positive Valence in Semantic Categorization

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Introduction

- In conversational interactions with people, AI LLMs are notoriously obsequious, sycophantic, cloyingly optimistic, excessively polite, fawning, pollyannish, unremittingly apologetic, saccharine, unctuous, complimentary to the point of absurdity, servile, subservient, inanely effusive, hyper-cheerful, ingratiating, consistently acquiescent, eager to please, overeager to please, obeisant, affectively monotonal, and in general tuned to a tone that can feel scrubbed of all potential friction to avoid user discomfort or even the slightest hint of unpleasantness.
- LLMs in voice chat are also well known to mirror user speech patterns, wishes and aims to create a comforting and supportive sense of alignment. Ironically, this mirroring is to some degree comforting exactly because it itself mirrors the human tendency to adopt the speech patterns and tones of human interlocutors.
- In this study, we investigated one aspect of this mirroring, namely the extent to which people acquiring new factual information in chats with LLMs encode the information in a way that mirrors the affectively positive valence of the context in which the information is learned.
- An obstacle to measuring consequences of LLM sycophancy is that because it is so widely recognized as an element of LLM chat interaction, and is so often derided, participants may be on their guard against revealing its effects. Here we used a semantic priming task that obliged participants to make speeded responses to categorize words and terms acquired in earlier conversations with LLMs. The speededness of the responding minimizes the potential for conscious and deliberate distortion in the direction of socially desirable responses.

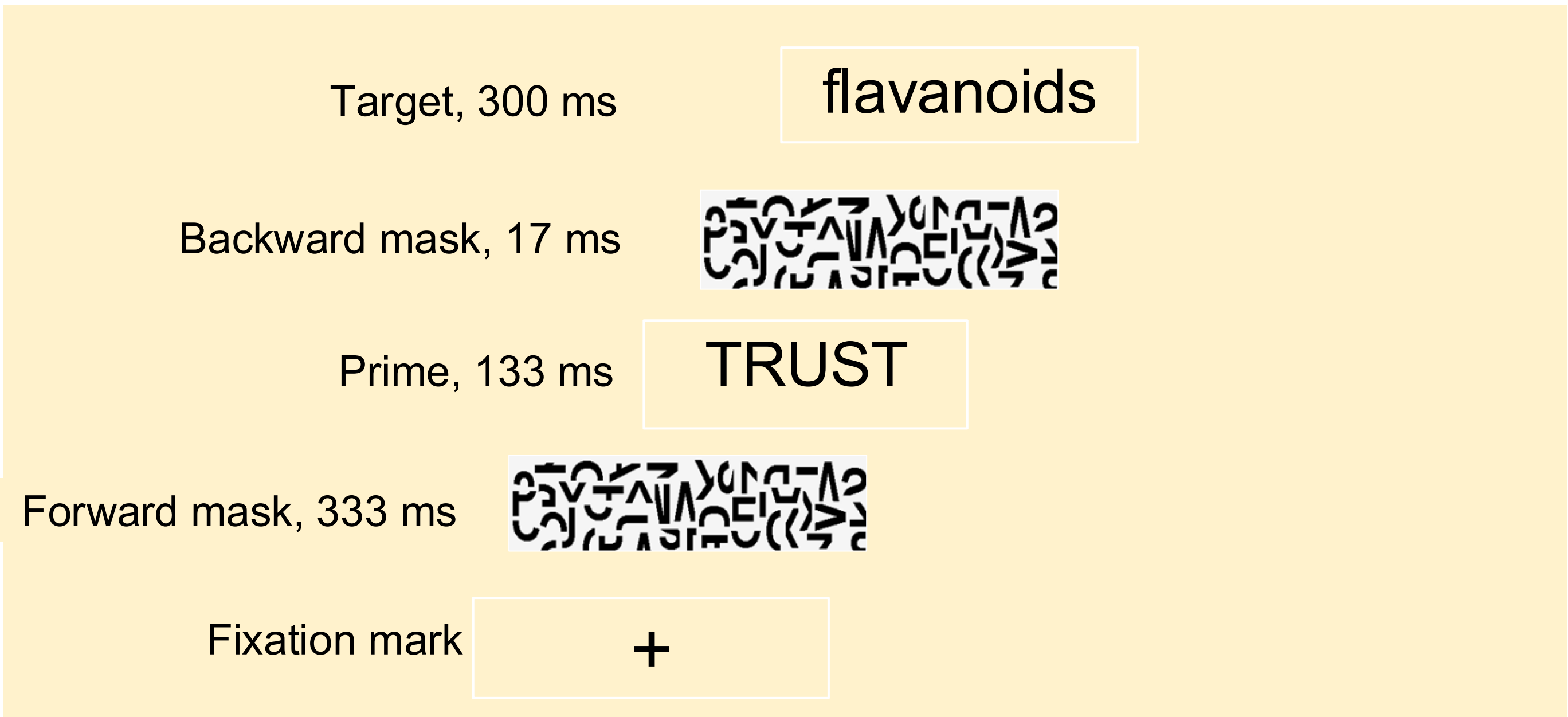
Method: Part 1, LLM chats

- Participants first did three sessions of guided interaction with OpenAI’s ChatGPT-5 (“Juniper” voice) on their own phones. The sessions were done on consecutive days (Monday, Tuesday, Wednesday), at times and in settings chosen for convenience by each individual participant.
- Each session lasted for approximately 25 minutes. Participants initiated a session by explaining that they were interested in the science and technology of two topics, rhubarb and plankton. They said that they had a list of 12 terms that they wanted to explore for each topic (see below). After discussing all the terms for one topic, they asked for a summary of each term. They then did the same for the other topic.
- Compliance with the instructions was assessed for each session by asking participants to take a screenshot of the topic summaries that followed each topic learning block.

Rhubarb terms		Plankton terms	
Anthaquinone	Victoria Variety	Heterotrophs	Cyanobacteria
Oxalic acid	Ascorbic Acid	Carbon sequestration	Mixotrophy
Perennial	Glycosides	Copepods	Silica frustules
Anthocyanins	Polygonaceae	Diatom	Calcium carbonate
Flavonoids	Rheum	Dinoflagellates	Nano eukaryotes
Petiole	Tannins	Autotrophs	Nitrogen fixation

Method: Part 2, semantic categorization

Trial sequence in the semantic categorization task



After their 3 consecutive days of GPT learning sessions, participants on the following day (Thursday) did a testing session built around trials structured as outlined above. The testing session lasted about 45 minutes, was done on in-classroom laptop workstations, and consisted of a practice phase and a data collection phase.

In the practice phase, participants familiarized themselves with the basic trial structure by making speeded lexical decision categorizations of a set of words and nonwords presented as targets. In anticipation of the data collection phase that would follow practice, the same masks that would appear in the data collection phase appeared in practice, but instead of a prime word, only a row of Xs (XXXXXX) was displayed. Participants were instructed to make their responses as quickly as possible and to respond progressively faster over the course of the 4 practice blocks.

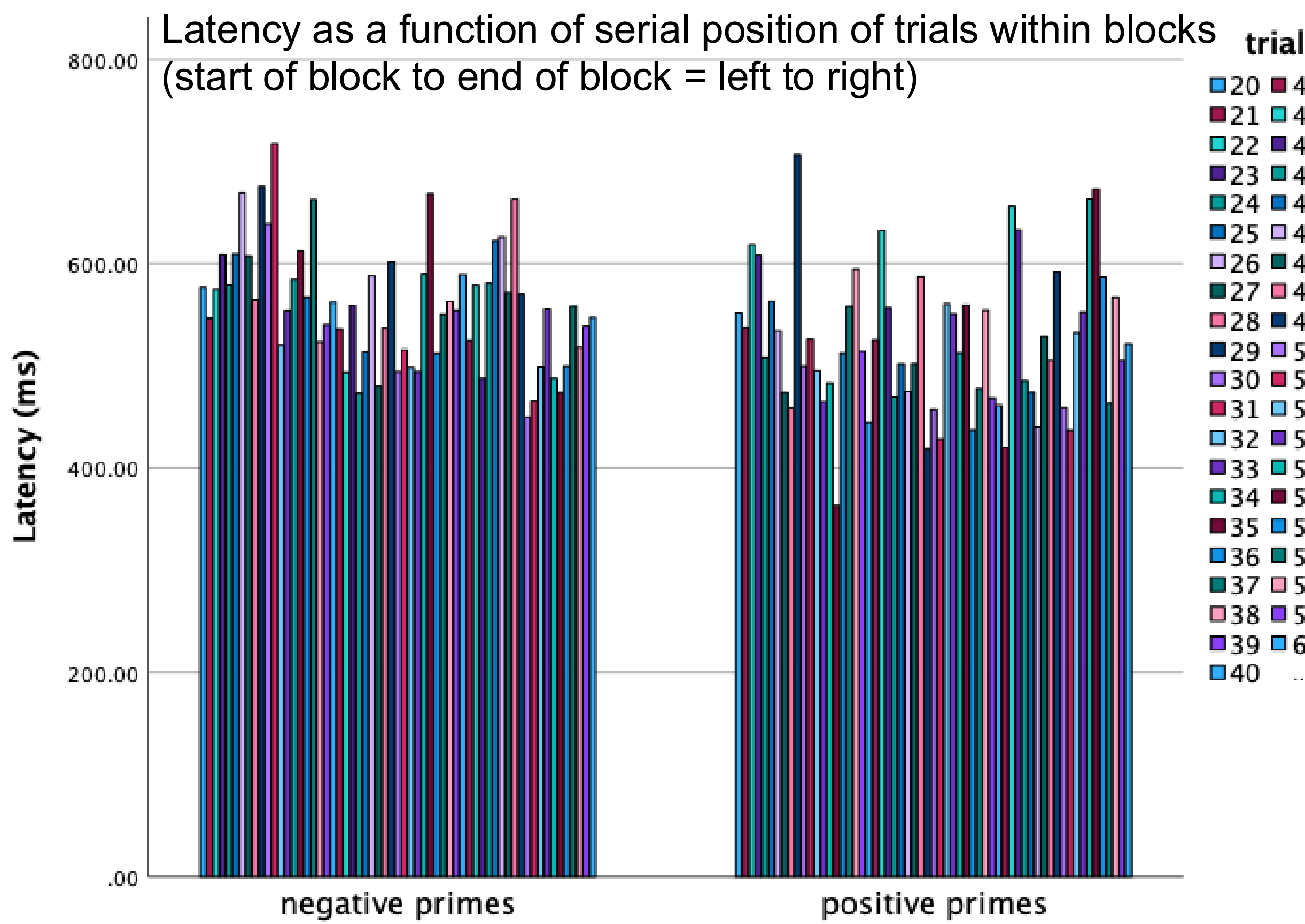
The data collection phase that followed consisted of 6 blocks of 80 trials each. In these blocks, the primes were words charged strongly with either positive or negative valence (e.g COZY, GRIEF), and the targets were the 12 rhubarb words/terms and 12 plankton words/terms from the learning-session prompts. The set of primes was drawn from the Bellezza, Greenwald, & Banaji (1986) ranked list (“Words high and low in pleasantness as rated by female and male college students”). Participants at the end of each block received feedback on their accuracy and reaction time.

Critically for the hypothesis test in this experiment, all of the primes in each block were either pleasant or unpleasant in meaning, thus creating an either positive or negative affective semantic environment within each block. Blocks were counterbalanced such that half the participants started with a pleasant-prime block and the other half with an unpleasant-prime block. The valence of primes then alternated for each counterbalancing group for the remaining blocks of data collection.

Within each trial, the primes were presented for a duration (133 ms) that under the masking conditions of the procedure permitted them to be fleetingly consciously perceptible for all of the participants we tested in a piloting phase of the study. The primes were presented in upper case and the targets in lower case to make clear which word participants were to respond to. Masks were constructed from letter fragments distributed across a rectangular area that spatially was slightly larger I the horizontal and vertical dimensions than the longest prime.

Results

- 28 participants were assessed as having complied with the LLM chat session instructions, and their data from the semantic categorization task were analyzed and are reported here.
- Data were analyzed by comparing latency and accuracy for positive-prime blocks versus negative prime blocks using a paired-samples *t* test. The first 20 trials of each 60-trial block were excluded from the analysis so as to mitigate “spillover” effects of valence from each previous opposite-valence block. For accuracy, slightly greater accuracy was observed in the positive prime blocks ($M = .70$; $SD = .13$) than in the negative prime blocks ($M = .68$; $SD = .14$), and the difference was marginally significant, $t(27) = 1.93$, $p = .068$. For latency—the dependent variable of focus—latency was about 20 ms faster in positive prime blocks ($M = 523.85$ ms; $SD = 163.56$) than in the negative prime blocks ($M = 554.82$ ms; $SD = 523.85$), and this difference too was marginally significant, $t(27) = 2.03$, $p = .053$.
- Of interest in the interpretation of the results is whether these (small) differences arose from the overall valence environment within each block—pleasant or unpleasant—as shaped by the uniform valence of the primes in each block, or whether the differences were driven by valence priming at the level of individual trials. Although the present methodology does not allow for a definitive answer, an inspection of latency over the course of the trials within blocks suggests that no pleasant or unpleasant valence “accrued” within each block as might be expected if blockwide valence were the primary determining factor:



Discussion

- The findings lend suggestive support to the notion that aspects of AI speech behavior are mirrored by and durably remain with human users who interact with LLMs like ChatGPT. More intriguingly, the findings imply that information acquired within the affectively charged envelope of LLM-human conversations may carry that charge beyond the conversation, and that such information is retrieved faster/more accurately in similarly charged contexts.
- A plausible alternative explanation to the present findings is that semantic categorization is generally faster/more accurate when primes are positively valenced (and that we would have observed a similar pattern of findings in a categorization task not preceded by LLM sessions). This alternative is straightforwardly testable.